

**CIE ACTIVITY 1:****SIMULATION-BASED STUDY OF EMBEDDED SYSTEM DESIGN USING PROTEUS**

Class:	TY	Semester:	V
Course Code:	2626PC12	Course Name:	Embedded Processors (EP)

Name of the Student:								Div.:					
Roll No.:					Student Code:								
ABC ID:													

Activity Description:

Design and *simulate* any one of the embedded-system problem statements given below by developing the corresponding system design and implementing the derived embedded application using either an **8051** or **PIC18F4550** microcontroller. *Develop* the required interfacing circuit, simulate the designed system using Proteus, and *analyze* the implementation based on the selection of peripherals, microcontroller selection criteria, and the obtained simulation results.

Learning Objectives

After completing this activity, students will be able to:

1. **Analyze** and select suitable microcontrollers and peripheral devices based on the requirements of the embedded system.
2. **interpret** the behavior and performance of the designed embedded system based on Proteus simulation results.
3. **Apply** the concepts of I/O port interfacing and built-in peripheral configuration of the selected microcontroller to realize the embedded system.

Instructions to the Students

1. **Select** any one embedded system design problem statement from the given list.
2. **Identify and justify** the selection of a suitable microcontroller and peripheral devices for the chosen problem statement.
3. **Develop** the system design, interfacing diagram, and algorithm for the selected embedded application.
4. **Implement and simulate** the designed embedded system using Proteus.
5. **Verify and analyze** the simulation results with respect to the expected system behavior.



6. **Present** the system design, interfacing diagram, simulation results, and analysis clearly for evaluation.

Assessment Criteria (Rubrics)

Final marks are to be scaled to 7.5

Performance Indicator	Excellent (5)	Very Good (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Preparation	Thoroughly reviewed theory and practical; ready to apply the design steps independently.	Reviewed most resources; shows good readiness to solve problems.	Reviewed partially; needs some support to apply concepts.	Limited preparation; struggles with key concepts.	Referred to some of the pre-class material shared.
Simulation and Problem Solving	Correctly simulates the designed using Proteus.	Mostly correct simulation with minor errors; demonstrates good logic.	Partial success with help; some coding or logic issues.	Struggles to simulate or relies heavily on support.	Unable to complete the simulation; major conceptual/code errors.
Result Interpretation	Clearly analyses the realization with respect to defined parameters and relates results to theory with insight.	Interprets most results accurately with a good explanation.	Basic interpretation with limited connection to theory.	Misinterprets results or explains vaguely.	No meaningful interpretation or explanation.

Problem Statement:



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Department of Electronics & Telecommunication Engineering

Embedded Processor: Continuous Internal Evaluation (CIE)

Academic Year: 2026-27

Solution:

1. System Design:

2. Peripheral and Microcontroller Selection:

Component	Purpose	Selection Criteria
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3. Interfacing Diagram:

4. Algorithm:

5. Uploaded URL:



6. Proteus Simulation Results: (Attach the screenshots of the simulation results)

7. Conclusion

Assessment:

Performance Indicator	Preparation (5)	Simulation and Problem Solving (5)	Result Interpretation (5)	Total (15)
Marks				

Signature of the Student

Signature of the Faculty